

Fabricating Accessible Computer Peripherals: An Ergonomic Approach to In-Bed Device Usage

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***Index Terms*—Ergonomic design, Computer peripherals, Human-computer interaction, Bedbound computing, Arduino controllers, Accessibility design**

I. INTRODUCTION

This project explores the development of ergonomic computer peripherals tailored for individuals who will be using it in nonstandard locations. We focus on designing a joystick-keypad combination, leveraging tools like Fusion 360 for 3D modeling, Tinkercad for circuit simulation, and Unity for software integration. Despite challenges such as design scaling issues and hardware fragility, the prototype highlights the potential for innovative accessibility solutions through iterative design, usability testing, and interdisciplinary methodologies.

II. MATERIALS AND METHODS

We conducted a thorough assessment to identify key challenges and requirements regarding using PC peripherals in nonstandard locations. Using the information we received, we brainstormed potential solutions and approaches to address the identified problems, and then got feedback on these solutions. We used the most promising idea, a joystick keypad combo, to make a low-fidelity design on paper and using clay. We then simulated the electrical circuits using Tinkercad (see Appendix H). We assembled the necessary electrical components based on the simulation on a breadboard and set up a primitive Unity scene with hardware integration (see Appendix J). We then designed 3D models using Autodesk Fusion 360 to create components for physical prototyping (see Appendix G). Then we used the Fusion360 model to 3D print (see Appendix E). We built and refined our Unity Scene with shooting gameplay, win and loss conditions. We soldered the electrical components so that they could fit in the 3D printed casing. Then we integrated all physical and digital components. Finally we conducted usability testing with end-users to evaluate performance, functionality, and user experience.

III. RESULTS

Once our controller was created in Fusion 360, we focused on updating the technical drawings, ensuring that they were accurately displayed on their respective sheets, with an exploded view of both peripherals with separate sheets depicting

their measurements with multiple views (See Appendix B). Unfortunately, as we'll mention later, the measurements were larger than expected which skewed the results when users were testing the controller. We used the System Usability Scale (SUS) questionnaire to gather the data taken from the user testing phase (See Appendix C). From those results, we determined that a majority of the participants were able to quickly pick up the controller and understand its functionality; most didn't find the controller to be cumbersome; and the intended functionality was well integrated into the controller scheme. However, the participants were split on whether they'd use the controller often. The results of this measure indicate that participants would seldom use our controller over the traditional mouse and keyboard or find alternative solutions to the problem. From the analysis gathered from the QFD, we can see that our controller ranks lower than our competitors (See Appendix D). From these results, we've determined that with further iteration, improvements, resources, and time our controller can compete properly with our competitors.

IV. CONCLUSION

As we assembled our controller, we faced many challenges, some of which were easy to fix while others were time-consuming and difficult to solve. First, the main electronics problem of our controller involved the unintentional breaking of some electronic components due to their fragile nature, the primary components to break often were push button pins and LEDs. To overcome this issue, we became extra careful while handling them and used tweezers and pliers to keep them steady during assembly; we also had backup push buttons and LEDs in case some components broke. Next, we encountered the most issues regarding the hardware/3D printed model. Unfortunately, the models of both peripherals were larger than needed to be, making them hard to use and control. This is a result of the measurements made within Fusion 360 being inaccurate to the measurements seen with our prototype and similar commercial devices; the internal circuitry and spacing within the controller weren't taken into consideration as thoroughly as they should have been. Lastly, the assembly of the controller took longer than expected which resulted in no room for iterative design and extensive testing; The projected time it would have taken to construct the controller would've

From these issues, we can learn from this first iteration of our device to improve the workflow for future iterations. To address the issue regarding time management, we should consider quick iterations near the beginning of development to create the best version of the controller earlier in the timeline; we can reduce the amount of troubleshooting for our device by solving these issues early on. Subsequently, we should consider the scale of our controller thoroughly to ensure it has enough room for internal components while offering an ergonomic design to users. Lastly, we recommend saving some time by 3D printing in bulk for quick prototyping and focusing on the timeline.

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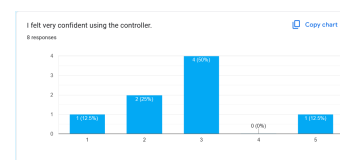
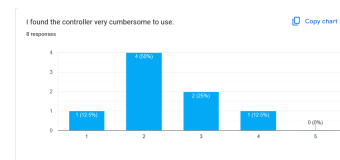
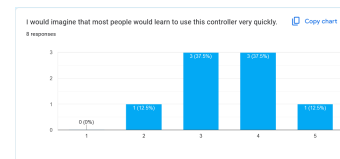
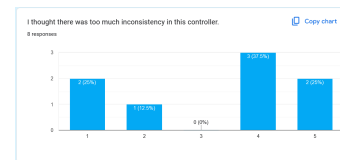
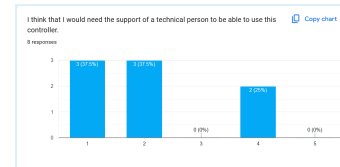
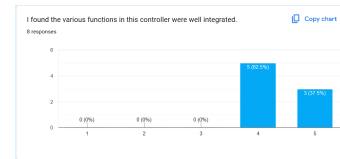
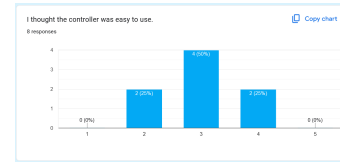
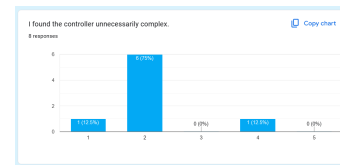
Wrote Results and Conclusion, created the joystick and keypad model in Fusion 360, constructed the physical assembly, worked on assembling the electronic components, and user testing questionnaire.

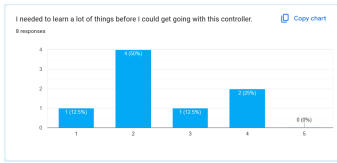
Sachin K

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Fig. 1. Project Timeline

Frequency	Number of Responses	Percentage
1	2	7.05%
2	2	7.05%
3	0	0.00%
4	2	7.05%
5	2	7.05%





APPENDIX D QFD

Link: <https://docs.google.com/spreadsheets/d/1kDc7n8EYmp0H-OdAZoQbTB1EtkL0mFlI5mQcojfSc1o/edit?usp=sharing>

APPENDIX E PROTOTYPE FABRICATION

Videos:
<https://youtu.be/RiJnDy-b1lM>
<https://youtu.be/m2qGsUJ82to>
https://youtu.be/TwZE_iQlsYg

APPENDIX F PROTOTYPE VIDEO

<https://youtu.be/7ciROaok7YI>

APPENDIX G TECHNICAL DRAWINGS

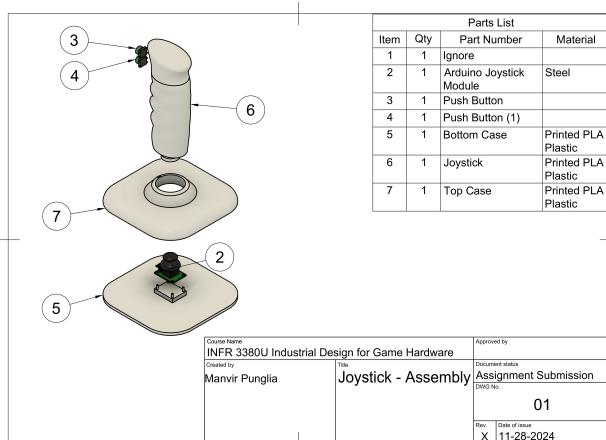


Fig. 2. Technical Drawing page 1

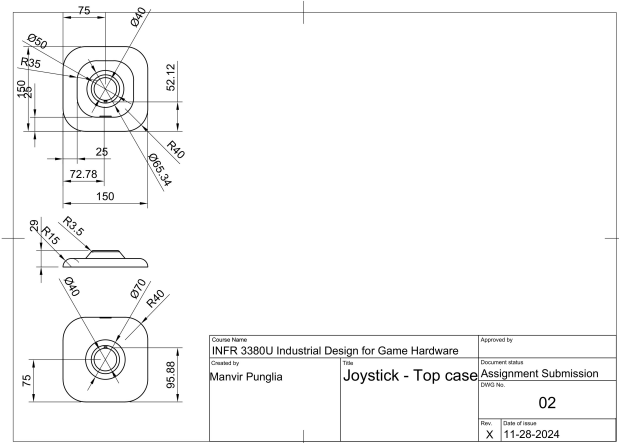


Fig. 3. Technical Drawing page 2

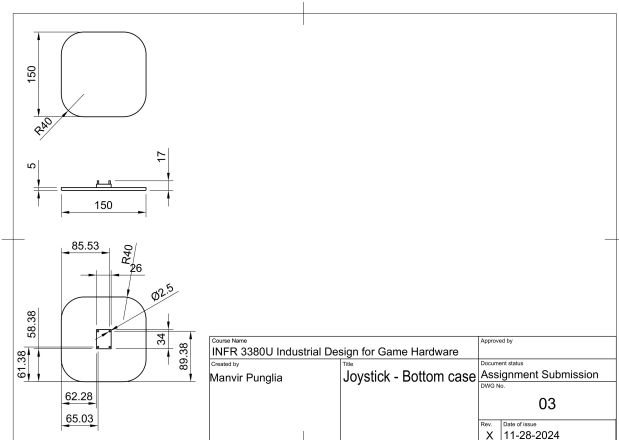


Fig. 4. Technical Drawing page 3

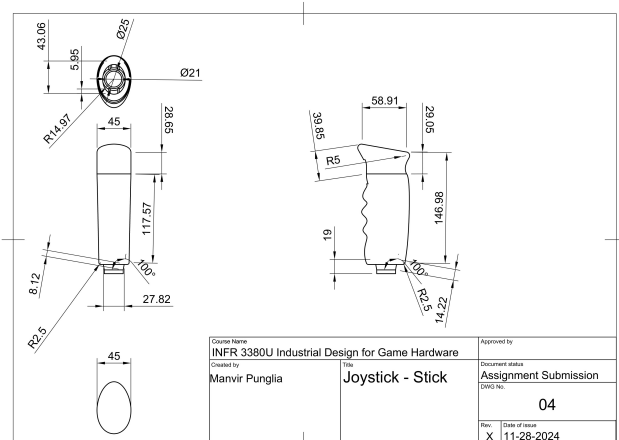


Fig. 5. Technical Drawing page 4

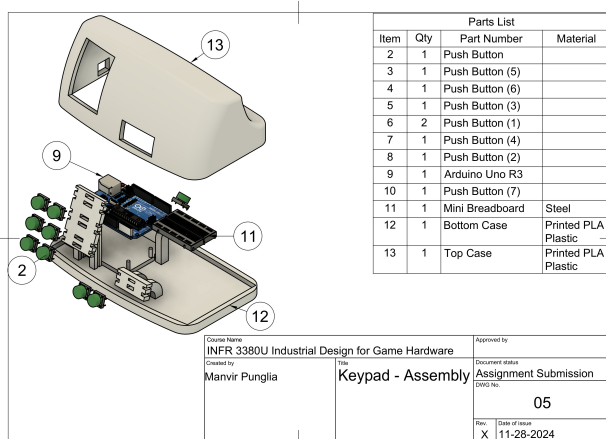


Fig. 6. Technical Drawing page 5

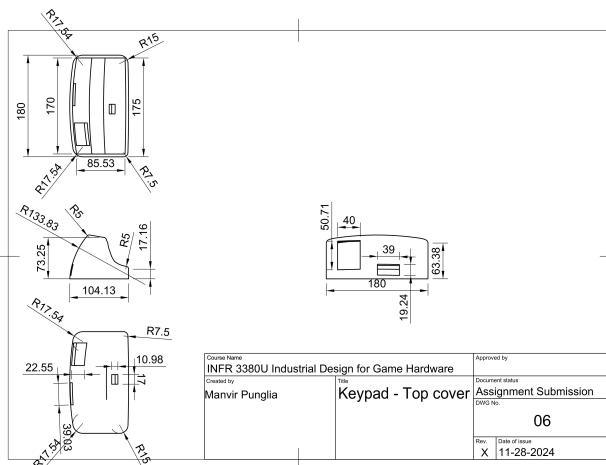


Fig. 7. Technical Drawing page 6

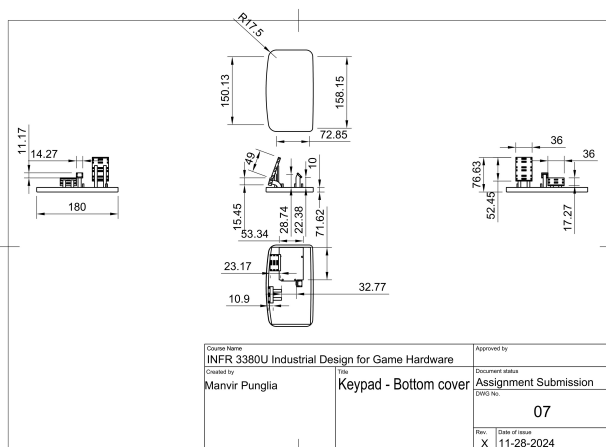


Fig. 8. Technical Drawing page 7

APPENDIX H ELECTRONIC SIMULATION

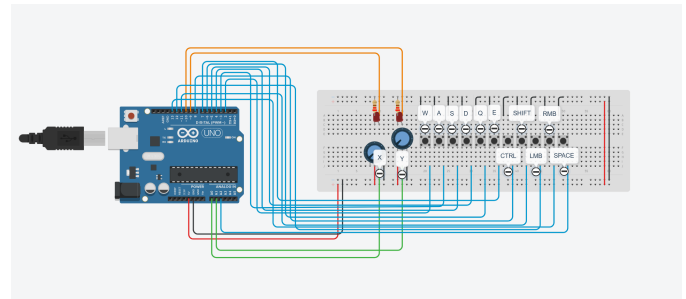


Fig. 9. The Electronic Schematics of our controller in Tinkercad

Video: <https://youtu.be/XIYwE0KpCkI>

APPENDIX I ASSEMBLY SIMULATION

Video: <https://youtu.be/FelWSb31p40>

APPENDIX J ELECTRONICS INTEGRATION

Video: <https://youtu.be/YN1L-BhYm7M>

APPENDIX K PARTS LIST

Parts List			
Item	Qty	Part Number	Material
1	1	Ignore	
2	1	Arduino Joystick Module	Steel
3	1	Push Button	
4	1	Push Button (1)	
5	1	Bottom Case	Printed PLA Plastic
6	1	Joystick	Printed PLA Plastic
7	1	Top Case	Printed PLA Plastic

Fig. 10. Joystick bill of materials

Parts List			
Item	Qty	Part Number	Material
2	1	Push Button	
3	1	Push Button (5)	
4	1	Push Button (6)	
5	1	Push Button (3)	
6	2	Push Button (1)	
7	1	Push Button (4)	
8	1	Push Button (2)	
9	1	Arduino Uno R3	
10	1	Push Button (7)	
11	1	Mini Breadboard	Steel
12	1	Bottom Case	Printed PLA Plastic
13	1	Top Case	Printed PLA Plastic

Fig. 11. Keypad bill of materials